

IOWA STATE UNIVERSITY
ELECTRICAL AND COMPUTER ENGINEERING

EE 2300

LAB 4 REPORT – SECOND-ORDER FILTERS

Date of Experiment: March 03, 2026

Date of Report Submission: March 09, 2026

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Sayon Saha

Introduction

In this lab, we started working with second-order filters. For experiment 1, we first built a first-order low pass active filter and then combined two of those filters in cascading configuration to create a second-order version of the filter. We measured the cutoff frequency; passband gain and quality factor for the circuit and compared them with our calculated values. For experiment 2, we built three different passive RLC filters, low-pass, high-pass and bandpass. Then we measured the passband gain, cutoff frequency and quality factor for each and compared them with our calculated values. We also used csv files from the loopinators to create MATLAB plots of magnitude and phase with two different resistors to compare them. For experiment three, we designed a low-pass Sallen-Key filter within certain specifications.

Experiment 1. Low-Q Low-Pass Active Filters

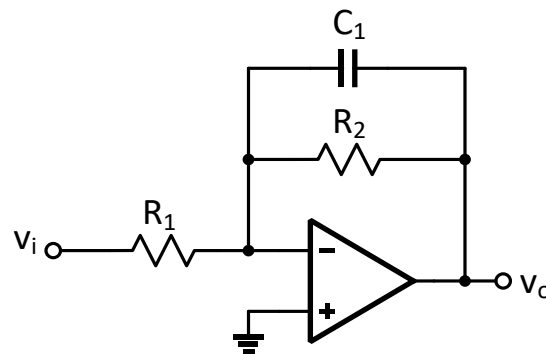


Figure 1-1. Schematic of first-order low-pass filter with $R_1 = 4.7 \text{ k}\Omega$, $R_2 = 10 \text{ k}\Omega$, and $C_1 = 10 \text{ nF}$.

The table below shows the calculated and measured cutoff frequency and pass-band gain for the first-order system.

	Calculated	Measured
Cutoff Frequency (kHz)	1.624	1.58
Passband Gain (dB)	6.48	6.34

The measured values I got for cutoff frequency and passband gain closely matched the calculated values. The calculated cutoff frequency was 1.624 kHz which was a bit higher than the measured value, and the calculated passband gain was 6.48 dB which was pretty close to the measured value. This slight difference can be attributed to component tolerances of the resistors and capacitor.

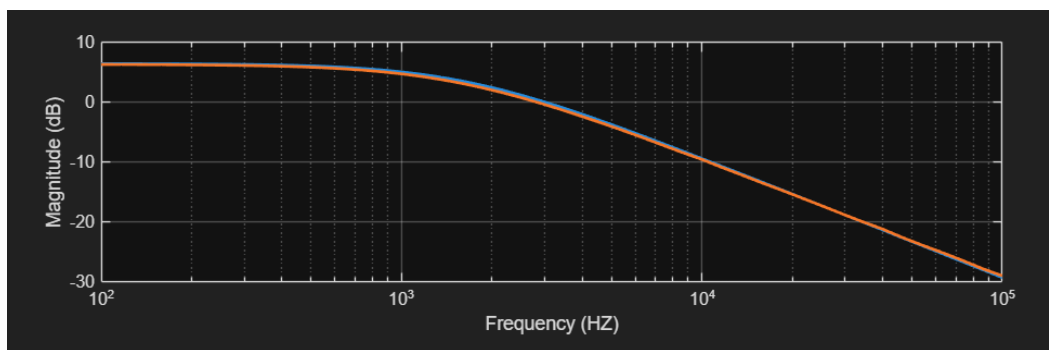


Figure 1-2. Plot of measured and calculated magnitude response.

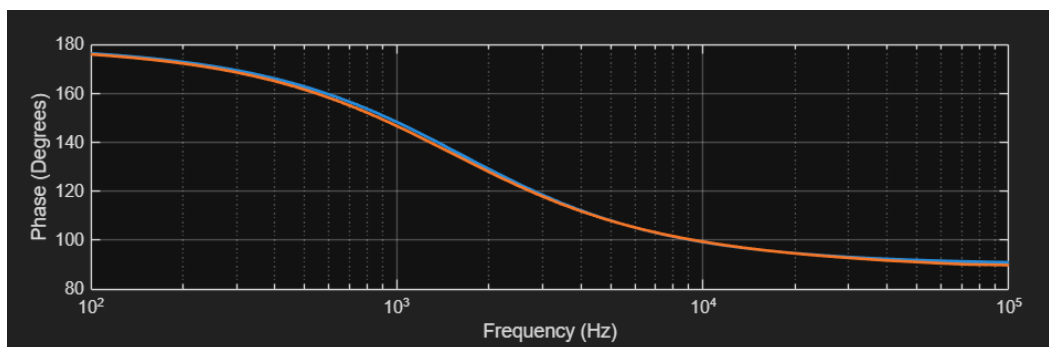


Figure 1-3. Plot of measured and calculated phase response.

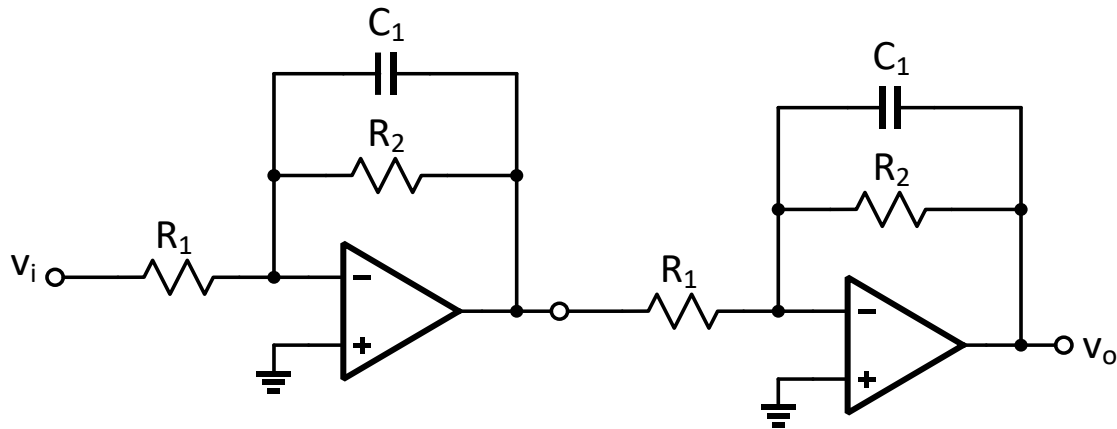


Figure 1-4. Schematic of second-order low-pass filter with $R_1 = 4.7 \text{ k}\Omega$, $R_2 = 10 \text{ k}\Omega$, and $C_1 = 10 \text{ nF}$.

The table below shows the calculated and measured cutoff frequency, pass-band gain, and quality factor for the second-order system.

	Calculated	Measured
Cutoff Frequency (kHz)	1.05	1.2
Passband Gain (dB)	13.1	12.94
Quality Factor	0.5	0.5

The measurements are reasonably close to the calculated values. The measured cutoff frequency was 1.2 kHz which was a bit higher than the calculated value of 1.05 kHz. The calculated passband gain was also slightly higher than the measured value. The quality factor was the same. The slight differences can be attributed to the component tolerance values again.

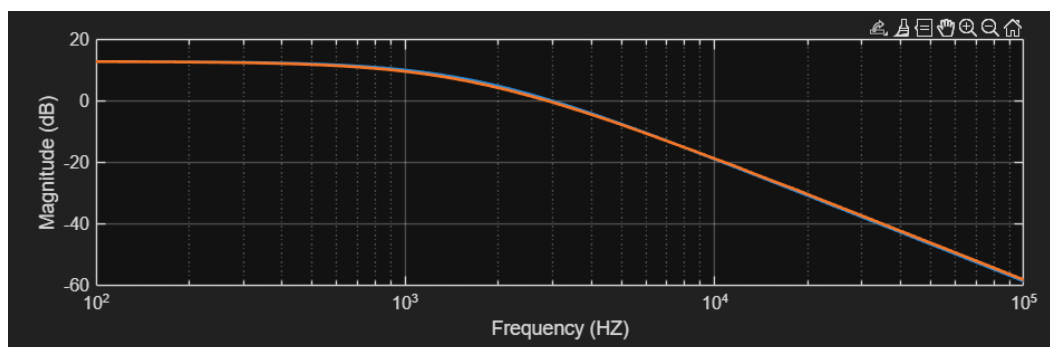


Figure 1-2. Plot of measured and calculated magnitude response.

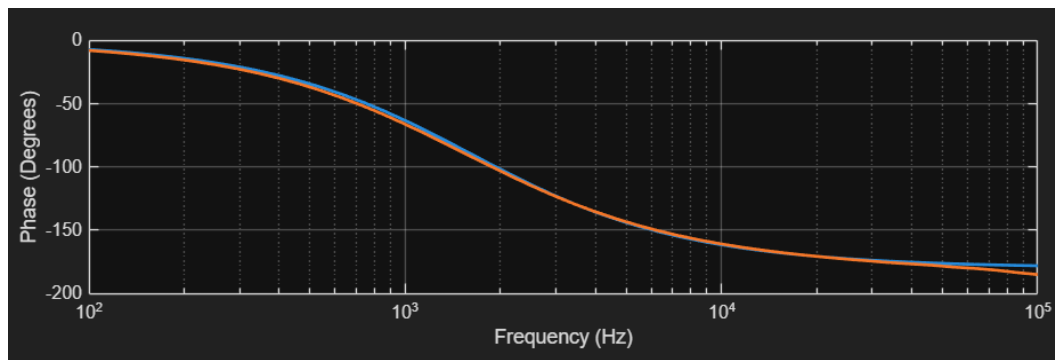


Figure 1-3. Plot of measured and calculated phase response.

Experiment 2. High-Q Second-Order RLC Filters

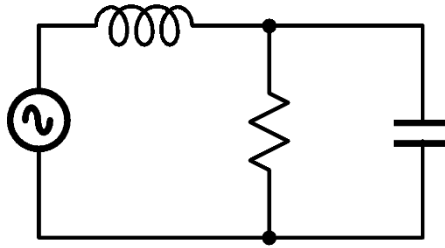


Figure 2-1. Schematic of the low-pass RLC filter with $L = 15 \text{ mH}$, $C = 0.1 \text{ } \mu\text{F}$ and R of $150 \text{ } \Omega$ or $1.5 \text{ k}\Omega$.

The table below shows the calculated and measured values for cutoff frequency, pass-band gain, and quality factor for the band-pass filter with $R = 1.5 \text{ k}\Omega$.

	Calculated	Measured
Cutoff Frequency (kHz)	6.31	6.1
Passband Gain (dB)	0	-0.26
Quality Factor	3.87	3.88

The measured values agree with the calculated values with slight deviations. The calculated cutoff frequency of 6.31 kHz was close to the measured value of 6.1 kHz. The passband gain ideally should be 0 dB, but we got -0.26 dB because of real components non-ideal characteristics. The quality factor was almost the same in the calculated and measured case. The slight deviations were because of non-ideal characteristics of the components.

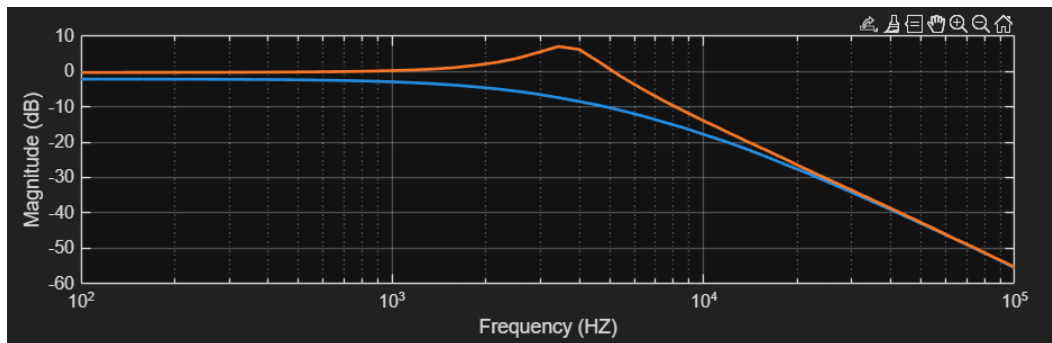


Figure 2-2. Plot of measured magnitude response of high-pass filter for $R = 150 \text{ } \Omega$ and $R = 1.5 \text{ k}\Omega$.

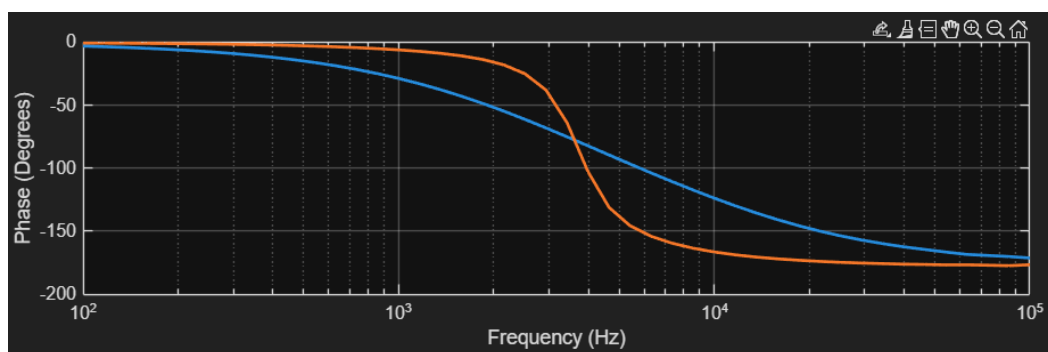


Figure 2-3. Plot of measured phase response of high-pass filter for $R = 150\ \Omega$ and $R = 1.5\ k\Omega$.

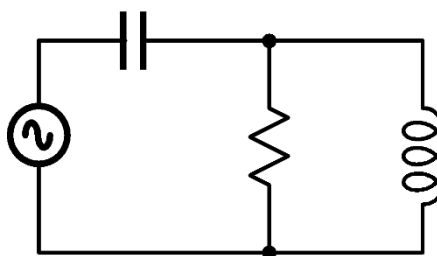


Figure 2-4. Schematic of the high-pass RLC filter with $L = 15\ mH$, $C = 0.1\ \mu F$ and R of $150\ \Omega$ or $1.5\ k\Omega$.

The table below shows the calculated and measured values for cutoff frequency, pass-band gain, and quality factor for the high-pass filter with $R = 1.5\ k\Omega$.

	Calculated	Measured
Cutoff Frequency (kHz)	2.7	2.5
Passband Gain (dB)	0	-0.205
Quality Factor	3.87	1.8

The calculated values are again close to the measured ones except the quality factor. The cutoff frequency and passband gain agree in both cases. But there is a noticeable difference between the calculated and the measured quality factor. This happened because of the inductor DCR. As we were measuring the output voltage across the inductor which brings the DCR into consideration in a physical circuit, and in ideal cases like calculating we didn't consider the DCR, major differences can be noticed in the quality factor values. Again, non-ideal characteristics only compound these differences.

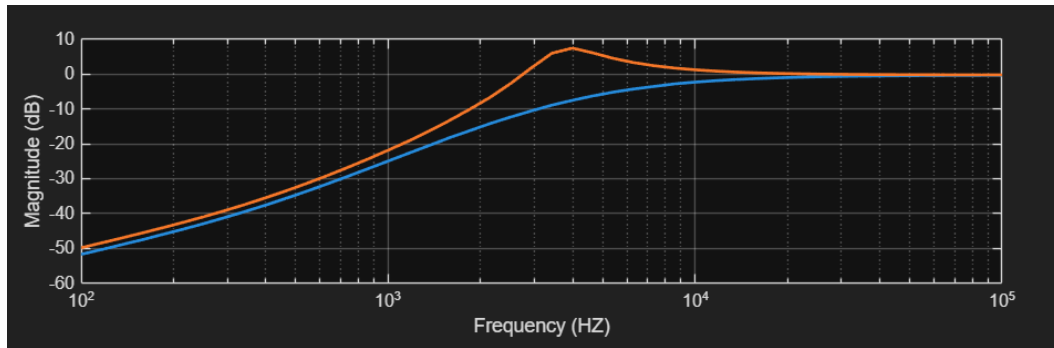


Figure 2-5. Plot of measured magnitude response of high-pass filter for $R = 150\ \Omega$ and $R = 1.5\ k\Omega$.

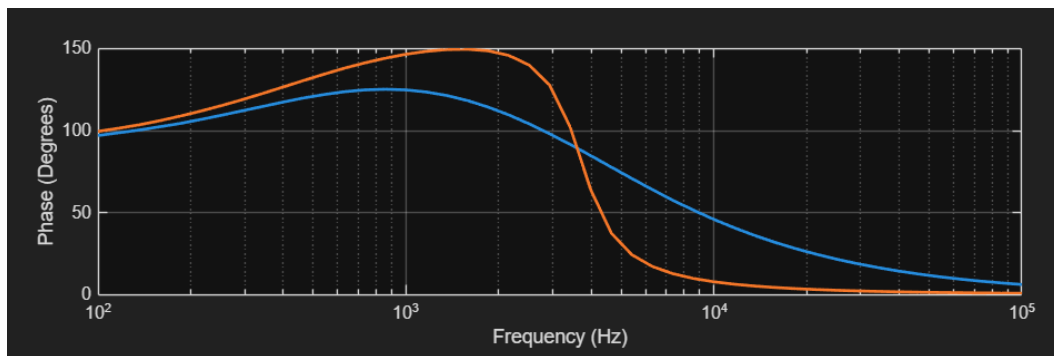


Figure 2-6. Plot of measured phase response of high-pass filter for $R = 150\ \Omega$ and $R = 1.5\ k\Omega$.

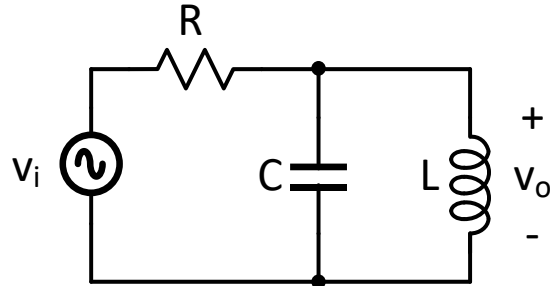


Figure 2-4. Schematic of the band-pass RLC filter with $L = 15 \text{ mH}$, $C = 0.1 \mu\text{F}$ and R of 150Ω or $1.5 \text{ k}\Omega$.

The table below shows the calculated and measured values for cutoff frequency, pass-band gain, and quality factor for the band-pass filter with $R = 1.5 \text{ k}\Omega$.

	Calculated	Measured
Cutoff Frequency (kHz)	3.28 and 5.03	3.28 and 5.03
Passband Gain (dB)	0	-5.093
Quality Factor	3.87	2.27

In this circuit all the measured values except the cutoff are significantly different from the calculated values. As we can see, our calculated passband gain value represents the ideal value of 0 dB, but when we measured it for the physical circuit, we got -5.093dB. This is a major difference, so is the difference between the calculated and measured quality factors. A major contributor to this is the DCR of the inductor not being considered when calculating using the ideal equations. The non-ideal characteristics also play a part.

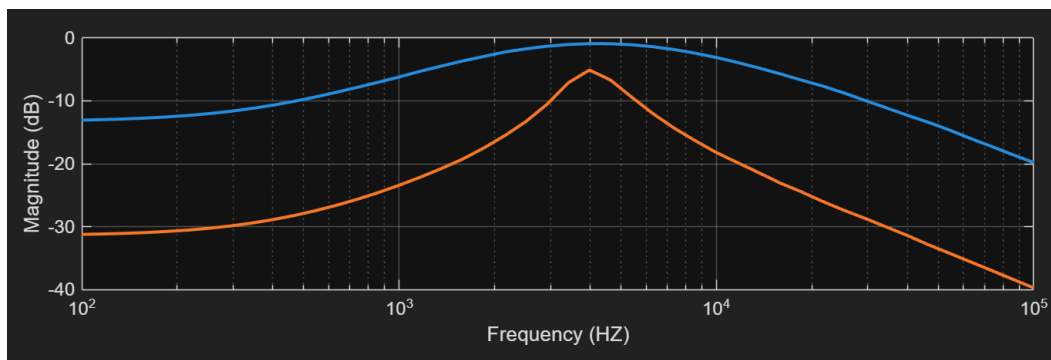


Figure 2-5. Plot of measured magnitude response of band-pass filter for $R = 150 \Omega$ and $R = 1.5 \text{ k}\Omega$.

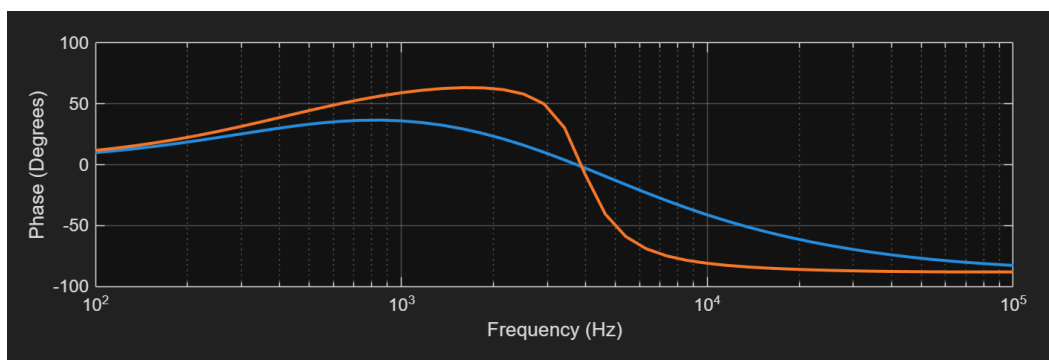


Figure 2-6. Plot of measured phase response of band-pass filter for $R = 150 \, \Omega$ and $R = 1.5 \, k\Omega$.

Experiment 3. Active Filter Design

To design the circuit, I first used the standard equations to find out n , which must be at least equal to $4Q^2$. There we got $n=2$, which is the ratio of the capacitors. Then I used n to figure out k and m , where m is the ratio of the resistor values. Then I fixed a capacitor (C_2) to $0.1\mu\text{F}$ and calculated the other component values from there.

Then I put two $0.1\mu\text{F}$ capacitors in parallel to get $C_1 = 0.2\mu\text{F}$, and as both the resistors required the same value, for each I put one $1\text{k}\Omega$, one 100Ω and one 22Ω resistor in series to get as close to 1125Ω as possible. I put them in the standard configuration for a Sallen-Key filter.

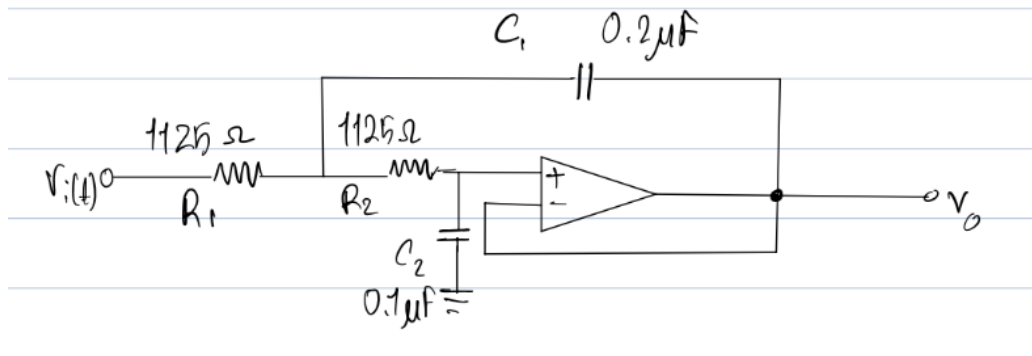


Figure 3-1. Schematic of the high-pass filter designed for this experiment.

The table below shows the simulated and measured characteristics of the design.

Parameter	Target	Simulated	Measured
Passband Gain	$1 \text{ V/V} \pm 5\%$	1	0.99
Cutoff Frequency	$1 \text{ kHz} \pm 10\%$	1.028	0.928
Quality Factor	$1/\sqrt{2} \pm 10\%$	0.707	0.712

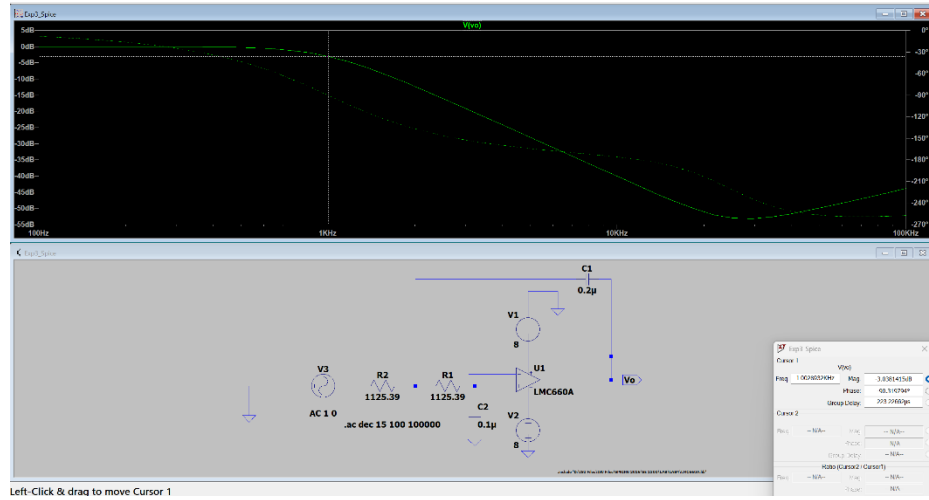


Figure 3-2. Screenshot of simulated frequency response (magnitude and phase) from LTSpice.

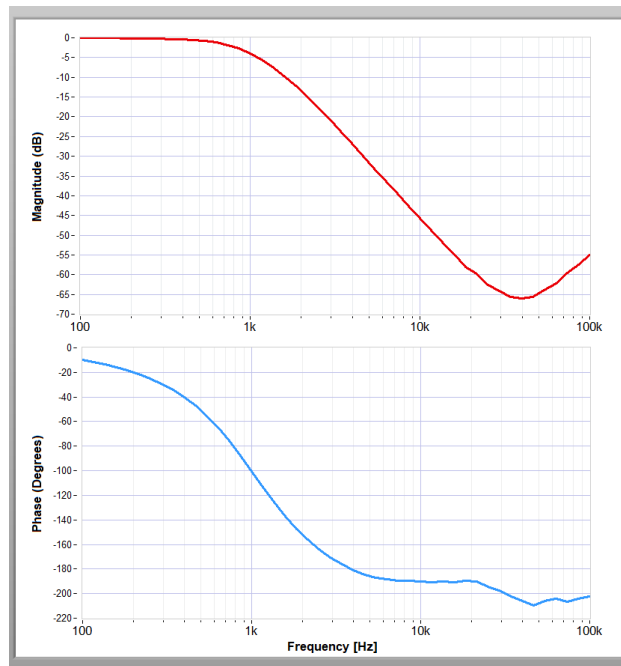


Figure 3-3. Measured frequency response (magnitude and phase).

Conclusion

In this lab, we didn't face much problem except the last parts of experiment 2, where our calculations portrayed very different quality factors and passband gain values from the ones we measured, especially for the bandpass filter. But later, we figured out that it was due to the DCR values of the inductor.

Calculations

Experiment 1:

$$R_f = C_1 \parallel R_2 = \frac{1/sC_1 \cdot R_2}{1/sC_1 + R_2} = \frac{R_2}{1 + sR_2C_1}$$

For an inverting op amp,

$$H(s) = -\frac{R_f}{R_1}$$

$$= -\frac{R_2}{R_1} \cdot \frac{1}{1 + sR_2C_1}$$

$$= -\frac{R_2}{R_1} \cdot \frac{1/R_2C_1}{s + 1/R_2C_1}$$

$$= -\frac{R_2}{R_1} \cdot \frac{1/\tau}{s + 1/\tau}, \text{ where } \tau = 1/R_2C_1$$

Measured,

$$R_1 = 4.64 \text{ k}\Omega$$

$$R_2 = 9.79 \text{ k}\Omega$$

$$C = 10.007 \text{ nF}$$

$$\text{so, } \omega_p = \frac{1}{R_2C_1}$$

$$\text{cutoff frequency, } f_c = f_p = \frac{\omega_p}{2\pi} = 1624.55 \text{ Hz}$$

$$\text{Passband gain, } G = -\frac{R_2}{R_1} = -2.11 \text{ V/V}$$

$$= 6.48 \text{ dB}$$

$$|H(j\omega)| = \frac{R_2}{R_1} \cdot \frac{\omega_c}{\sqrt{\omega_c^2 + \omega^2}}$$

$$\angle H(j\omega) = \pi - \tan^{-1} \left(\frac{\omega}{\omega_c} \right)$$

Cascading circuit,

$$H_2(s) = H_1(s) \cdot H_1(s)$$

$$= \left(\frac{R_2}{R_1} \right)^2 \left(\frac{1/R_2 C_1}{s + 1/R_2 C_1} \right)^2$$

$$|H_2(j\omega)| = \left(\frac{R_2}{R_1} \right)^2 \frac{1}{1 + (\omega/\omega_c)^2}$$

$$\angle H_2(j\omega) = 2\pi - 2 \tan^{-1}(\omega R_2 C_1)$$

From the standard equation,

$$G_0 = \left(\frac{R_2}{R_1} \right)^2 = 4.53 \text{ V/V} = 13.1 \text{ dB}$$

$$f_c = \frac{\omega_c}{2\pi} = \frac{(1/R_2 C_1)}{2\pi} = 1.05 \text{ kHz}$$

$$\frac{\omega_0}{Q} = \frac{2}{R_2 C_1} \Rightarrow Q = \frac{1}{2}$$

Experiment 02:

For the first circuit,

$$\text{so, } \omega_0 = \frac{1}{\sqrt{LC}} \\ = 25819.88 \text{ rad/s}^{-1}$$

In the circuit, CIR
 $\text{so, } Z_{CR} = \frac{\frac{1}{sC} \cdot R}{\frac{1}{sC} + R} = \frac{R}{1+sRC}$

$$Q = R\sqrt{C/L} \\ = 3.87$$

Applying voltage divider,

$$\begin{aligned} \frac{V_o}{V_i} = H(s) &= \frac{\frac{R}{1+sRC}}{\frac{R}{1+sRC} + sL} \\ &= \frac{R}{R + sL(1+sRC)} \\ &= \frac{R}{s^2 RLC + sL + R} \\ &= \frac{\frac{1}{LC}}{s^2 + s\left(\frac{1}{RC}\right) + \left(\frac{1}{LC}\right)} \end{aligned}$$

now, comparing with the standard equation,

$$H(s) = G_0 \frac{\omega_0^2}{s^2 + \left(\frac{\omega_0}{Q}\right)s + \omega_0^2}$$

$$f_0 = \frac{\omega_0}{2\pi} = \frac{\frac{1}{\sqrt{LC}}}{2\pi} = \boxed{\frac{1}{2\pi\sqrt{LC}}}$$

$$Q = \omega_0 RC = \frac{1}{\sqrt{LC}} \cdot RC = \boxed{R\sqrt{C/L}}$$

$$\boxed{G_0 = 1}, \text{ because of DC gain}$$

Passband gain $H(0) = 1 \text{ V/V} = 0 \text{ dB}$

$$\text{Cutoff frequency, } f_c = \frac{\omega_c}{2\pi} = \frac{\omega_0 \sqrt{1 - 1/2Q^2} + \sqrt{1 + (1 - 1/2Q)^2}}{2\pi}$$

$$= 6309.45 \text{ Hz}$$

For the high pass filter circuit,

$$Q = 3.87$$

Here, $R \parallel L$,

$$\text{so, } Z_{RL} = \frac{sL \cdot R}{sL + R}$$

now, voltage divider,

$$H(s) = \frac{\frac{sRL}{sL + R}}{\frac{1}{sC} + \frac{sRL}{sL + R}}$$

$$= \frac{sRL}{\frac{1}{sC}(sL + R) + sRL}$$

$$= \frac{sRL}{\frac{L}{C} + \frac{R}{sC} + sRL}$$

$$= \frac{s^2 RL}{s^2 RL + s \frac{L}{C} + \frac{R}{C}}$$

$$= \frac{s^2}{s^2 + s\left(\frac{1}{RC}\right) + \frac{1}{LC}}$$

now, comparing with the standard form,

$$H(s)_{HP} = G_0 \frac{s^2}{s^2 + s\left(\frac{\omega_0}{Q}\right) + \omega_0^2}$$

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1/\sqrt{LC}}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$$

$$Q = \omega_0 RC = R\sqrt{\frac{C}{L}}$$

$$G_0 = 1 \text{ (High frequency gain)}$$

Passband gain, $H(\infty) = 1 \text{ V/V} = 0 \text{ dB}$

$$\text{Cutoff frequency, } f_c = \frac{\omega_c}{2\pi} = \frac{\omega_0 \sqrt{1 - 1/2Q^2} + \sqrt{1 + (1 - 1/2Q)^2}}{2\pi}$$

$$= 2.7 \text{ kHz}$$

Here,

$$s0, Z_L = \frac{1/sC - sL}{1/sC + sL} = \frac{1/C - s^2 LC}{1 + s^2 LC}$$

applying voltage divider,

$$H(s) = \frac{\frac{sL}{1+s^2LC}}{R + \frac{sL}{1+s^2LC}}$$

$$= \frac{sL}{R(1+s^2LC) + sL}$$

$$= \frac{s}{s^2 RC + s + R/L}$$

$$= \frac{s \cdot 1/kC}{s^2 + s(1/kC) + 1/LC}$$

now,

comparing with the standard form,

$$H(s)_{BP} = G_0 \frac{s \cdot \frac{\omega_0}{Q}}{s^2 + s\left(\frac{\omega_0}{Q}\right) + \omega_0^2}$$

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$$

$$Q = R\sqrt{\frac{C}{L}}$$

$$G_0 = 1$$

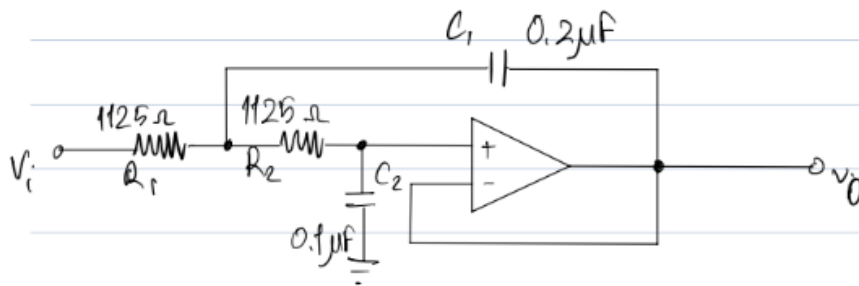
For the passband filter,

$$H(s) = \frac{s \cdot 1/kC}{s^2 + s(1/kC) + 1/LC}$$

$$Q = 3.87$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 25819.89 \text{ rad/s}^{-1}$$

$$\text{passband gain, } |H(j\omega_0)| = \frac{1/k\omega_0}{\sqrt{1 - \left(\frac{\omega}{\omega_0}\right)^2 + \left(\frac{\omega}{Q\omega_0}\right)^2}} = 0 \text{ dB}$$

Experiment 03:

We know,

$$\frac{C_1}{C_2} = n \geq 4Q^2$$

Taking $Q = 1/\sqrt{2}$,

$$n = 2$$

$$\text{so, } C_1 = 2C_2$$

now,

$$K = \frac{C_1/C_2}{2Q^2} - 1$$

$$= 2 - 1$$

$$= 1$$

$$\text{so, } m = K + \sqrt{K^2 - 1}$$

$$= 1$$

which means,

$$\frac{R_1}{R_2} = 1$$

$$\Rightarrow R_1 = R_2$$

Let's take, $C_2 = 0.1 \mu\text{F}$ from the lab kit

$$\text{so, } C_1 = 0.2 \mu\text{F}$$

$$\text{and at } f_0 = f_c = 1 \text{ kHz} \rightarrow \omega_0 = 6283.2 \text{ rad/s}^1$$

$$\text{so, } R_2 = \frac{1}{\omega_0 C_2 \sqrt{mn}} = 1125.35 \Omega = R_1$$

so, we need,

$$C_1 = 0.2 \mu\text{F}, C_2 = 0.1 \mu\text{F}, R_1 = R_2 \approx 1125 \Omega$$

For C_1 , we can have two $0.1 \mu\text{F}$ capacitors in parallel

For R_1 & R_2 , we can use $1 \text{ k}\Omega$, 100Ω and 22Ω resistors in series to achieve close to 1125Ω